TSMC (TSM): Hold, Valuation Concerns

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The indispensable foundry of the global digital economy, priced for perfection. At $419, the market prices 28.7% annual revenue growth for five years, above even our 21% bull case. Our DCF sees $243 base case, 42% downside to a $200 bear and $293 bull.

Hold, Valuation Concerns	$243	$419.06	-42%
RATING	BASE TARGET (BLENDED $245)	SPOT (OCT 15, 2026)	DOWNSIDE TO BASE

$200 / $293	9.8% → 8.7%	$46.7B (+41%)	68%	$55.3B	1.37 → 1.05
BEAR / BULL TARGETS	LIVE WACC (Y1→Y5)	Q3 REVENUE YOY	HPC REVENUE MIX	NET CASH	BETA DECAY

Executive summary

We initiate coverage of Taiwan Semiconductor Manufacturing Company with a **Hold, Valuation Concerns** and a $243 base-case target (blended $245). The operating story is as close to a monopoly as the technology industry offers: 79% of wafer revenue from 7-nanometer and smaller geometries, HPC at 68% of the revenue mix growing 22% sequentially, the 2-nanometer node ramping from 3% to 8% of wafer revenue in a single quarter, and Q4 guidance of $52.3 to $54.3 billion that management raised with unusual conviction.

TSMC is the picks-and-shovels winner of every major technology wave. It rode the smartphone boom of the 2010s, it is riding the AI infrastructure boom now, and it is positioned to ride the robotics boom next. Each wave demands more advanced silicon, and there is exactly one reliable supplier of leading-edge logic. That structural position is the core of the long-term bull case, and we do not dispute it.

The valuation, however, has run ahead of even that exceptional story. At $419, our reverse DCF shows the market prices 28.7% annual revenue growth for five years, above our 21% bull case. Our five-year detailed EBIT-bridge DCF (9% / 15% / 21% revenue CAGRs, 44 to 56% EBIT margins, 30 to 36% capex intensity reflecting a foundry that must build fabs) yields Bear $200, Base $243, Bull $293 at a live 9.8% Year 1 WACC decaying to 8.7% (1.37 beta to a 1.05 mature semi peer, 5.08% 10-year, 3.53% implied ERP). The business is superb. The equity is priced for an outcome with almost no precedent at a $2.2 trillion market cap.

> **Our read:** TSMC captures every major tech trend, and the robotics wave could extend the growth runway well past 2030. But the current price already assumes the AI boom never decelerates and the next wave arrives on schedule. With 28.7% implied growth versus our 21% bull case, there is no margin of safety. We would turn constructive below $270, where the base case begins to offer compensation for Taiwan concentration risk.

Base-case assumptions

INPUT	BEAR	BASE	BULL
5-yr revenue CAGR (FY26-30)	9%	15%	21%
2030 revenue ($B)	248	308	374
EBIT margin path	44% → 48%	48% → 52%	50% → 56%
Capex (% of revenue)	30%	33%	36%
Terminal growth	3.0%	3.0%	3.0%
Target per share	**$200**	**$243**	**$293**

WACC 9.8% Year 1 decaying to 8.7% Year 5 (spot 10Y 5.08% + beta 1.37→1.05 x 3.53% implied ERP). Gordon terminal. Net cash $55.3B; 5,186.5M shares. FY2026E revenue $176B (Q1-Q3 actual + Q4 guide midpoint).

The wave rider: smartphones, AI, and next robotics

TSMC's most underappreciated asset is its position across technology cycles. In the 2010s, the smartphone boom made TSMC indispensable: every flagship phone ran on its leading nodes, and the company compounded revenue at roughly 10% annually through the cycle. In the 2020s, the AI infrastructure boom has been even more lucrative. HPC now commands 68% of revenue, up from a minority share five years ago, and hyperscaler demand for 3nm and 5nm capacity has pushed utilization to maximum.

The next wave is likely robotics. Humanoid robots, autonomous systems, and edge AI inference all require advanced logic at massive scale, and TSMC will manufacture effectively all of it. The company's CoWoS advanced packaging capacity, currently the binding constraint on AI accelerator shipments, is directly transferable to the complex multi-die packages that robotics platforms will need. In our view, investors who frame TSMC as an "AI stock" are missing the larger pattern: it is a picks-and-shovels monopoly on compute itself, and compute demand has grown through every cycle.

This structural argument is the foundation of our bull case. A 21% revenue CAGR to $374 billion by 2030 assumes the AI boom sustains and the robotics ramp begins contributing in the outer years. Even that optimistic path only supports $293 per share at our WACC, 30% below spot. The thesis is right; the price already reflects it several times over.

Competitive moat

TSMC's moat is arguably the deepest in global technology. Intel Foundry and Samsung continue to struggle with yields on sub-5nm equivalent nodes, leaving TSMC as the sole reliable supplier of bleeding-edge logic. Switching costs are extreme: fabless designers spend billions optimizing architectures for TSMC's process design kits, and requalifying a design on a competitor's process takes years. Scale reinforces the lead: $40.6 billion of capex in the first nine months of 2026 alone funds an R&D flywheel that sub-scale peers cannot replicate. Advanced packaging (CoWoS) extends the moat from front-end lithography to back-end integration, a second bottleneck the company controls.

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Bull case: $293 AI HYPER-CYCLE + ROBOTICS

- **21% revenue CAGR to $374B by 2030:** hyperscaler AI buildouts sustain, N2 ramp executes cleanly, CoWoS capacity aligns with demand, and early robotics silicon orders appear in 2029-2030.
- **Margin expansion to 56%:** N2 yields mature faster than expected, pricing power on leading nodes offsets Arizona fab cost premiums, and operating leverage returns as depreciation normalizes.
- **36% capex funds the next moat:** heavy reinvestment extends the technology lead; mature FCF margins approach 30% as the 2030s begin.
- **Geopolitical premium fades:** Arizona and Japan fabs derisk Taiwan concentration, and the market awards a lower risk premium.

Even this optimistic path is worth $293, 30% below spot. Perfection is not enough at $419.

Bear case: $200 AI CAPEX DIGESTION

- **9% revenue CAGR:** AI infrastructure spending normalizes after the 2024-2026 hyper-build, hyperscalers digest capacity, and the robotics wave arrives later than hoped.
- **N2 disappointment:** yield curve proves steeper than the 3nm transition, gross margins compress below 62%, and early depreciation drags operating margins to 44%.
- **Overseas fab drag:** Arizona cost premiums prove structural, not transitory, permanently impairing consolidated margins by 200-300 bps.
- **Taiwan risk repriced:** cross-strait tensions widen the required risk premium, compressing the multiple the market will pay for the cash flows.

Verdict

We rate TSM Hold, Valuation Concerns. This is not a Sell: the business is exceptional, the moat is the deepest in technology, and the wave-rider thesis (smartphones to AI to robotics) gives TSMC a longer growth runway than any semiconductor company in history. But at $419 the equity prices 28.7% annual growth, a pace that exceeds our bull case and has almost no precedent at this scale. Overvalued plus healthy business equals Hold, not Sell. We would turn Buyers below $270.

What would change our mind

- **Price:** a decline below $270 would bring the base case into a favorable risk-reward zone; below $220 the bear case itself offers value.
- **Sustained 25%+ growth:** three consecutive years above 25% revenue growth with stable margins would support raising the bull case toward 25% CAGR.
- **Robotics inflection:** credible large-scale robotics silicon orders (tens of billions in annual revenue) would extend the terminal growth assumption.
- **Multiple compression without fundamental cause:** a geopolitical scare that leaves operations untouched would be a buying opportunity, not a thesis break.

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TSM Q3 2026 earnings: AI top-line velocity vs. 2-nanometer ramp frictions

Executive summary

Total revenue printed at $46.72 billion for Q3 2026, up 41.2% year over year and beating the upper end of management's guidance. Growth was overwhelmingly powered by High-Performance Computing demand, which surged 22% sequentially to 68% of platform revenue. Gross margin compressed to 66.4% from 67.7% in Q2 on early 2-nanometer depreciation and yield-curve costs. Management raised Q4 revenue guidance to $52.3-54.3 billion, a midpoint of $53.3 billion that signals extreme confidence in the order backlog. Net profit reached NT$763.06 billion, up 68.7% year over year, demonstrating profound operating leverage even through the node transition.

Bull case

- **Advanced node dominance, 79% of wafer revenue:** 7-nanometer and smaller geometries command nearly four-fifths of wafer revenue, cementing pricing power in AI hardware.
- **Q4 guide midpoint $53.3B:** the aggressive sequential leap proves deep visibility into sustained hyperscaler deployments.
- **N2 adoption velocity:** 2-nanometer leapt from 3% to 8% of wafer revenue in one quarter, the fastest leading-node ramp in company history.
- **CoWoS bottleneck clearing:** advanced packaging capacity additions translate directly to sequential revenue beats as AI accelerator shipments unclog.

Bear case

- **N2 margin drag, 66.4% gross margin:** early-stage depreciation and yield costs will pressure margins through 2027; this is the standard cost of node transitions, but it is real.
- **Extreme capital intensity, $13.84B Q3 capex:** relentless fab expansion restricts free cash flow conversion; 9M 2026 capex of $40.64B annualizes above $54B.
- **Customer concentration:** HPC at 68% means a synchronized hyperscaler capex pause would hit the revenue base directly.
- **Taiwan concentration:** the most advanced capacity remains in Taiwan; Arizona fabs carry structural cost premiums.

Verdict: Bullish on the quarter

The quarter confirms the AI infrastructure boom remains in its most aggressive phase. Top-line velocity and the raised guide eclipse the frictional costs of the N2 transition. We are bullish on the fundamentals and concerned only on valuation.

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Key themes

1. HPC is the definitive growth engine ACCELERATING

HPC drove the bulk of the $46.72 billion quarter and reached 68% of the platform mix, up 22% sequentially. Hyperscaler AI buildouts keep 3nm (33% of revenue) and 5nm (28%) utilization at maximum. This segment insulates TSMC from lingering softness in consumer electronics and concentrates the company's fortunes, for better and worse, on the AI capex cycle.

2. CoWoS expansion clears the AI bottleneck ACCELERATING NEW

TSMC is aggressively scaling Chip-on-Wafer-on-Substrate capacity for NVIDIA, AMD, and other accelerator designers. Advanced packaging has become the binding constraint on AI server shipments, and each increment of CoWoS capacity translates directly to revenue. This extends the moat from lithography to system-level integration.

3. Leading-edge adoption velocity ACCELERATING

N2 moving from 3% to 8% of wafer revenue in one quarter is extraordinary. Customers are migrating to the most advanced nodes faster than in any prior transition, which secures long-term lock-in: once a design tapes out on N2, it stays on TSMC for its commercial life.

4. N2 ramp frictions pressure margins DECELERATING NEW

Gross margin fell to 66.4% on early N2 depreciation and yield-curve costs. Management guided Q4 gross margin to 65-67%, signaling the drag persists near term. We model this as cyclical, not structural: every leading-node transition in TSMC's history has followed the same pattern of early dilution followed by margin recovery at scale.

5. Capital intensity suppresses near-term FCF STABLE

$40.64 billion of capex through nine months annualizes to a record year, and the $60-64 billion full-year range confirms maximum-velocity reinvestment. Q3 free cash flow of TWD 476.56 billion despite these outlays highlights supreme underlying cash generation. In our DCF we hold capex at 30-36% of revenue across scenarios, appropriate for a foundry that must build fabs to grow.

6. Overseas fab cost premiums STABLE NEW

Arizona and Japan facilities carry structural cost premiums versus Taiwan operations: higher labor and construction costs, smaller scale, and duplicative overhead. These satisfy geopolitical diversification mandates from major clients, but investors should expect a permanent 100-200 bps drag on consolidated margins versus an all-Taiwan footprint.

7. The robotics option ACCELERATING NEW

Humanoid robotics and physical AI represent the next compute wave, and TSMC will manufacture effectively all leading-edge robotics silicon. This is not yet in revenue, but it extends the terminal growth runway and supports our 3.0% terminal growth rate. Any acceleration here is upside to all three scenarios.

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Other KPIs

METRIC	VALUE
Market cap (at $419.06)	$2.17T
Enterprise value	$2.12T
FY2025 revenue / YoY	$121.4B / +37.6%
FY2025 net income / margin	$54.0B / 44.5%
FY2025 operating income / margin	$61.2B / 50.4%
FY2025 operating cash flow	$72.5B
FY2025 free cash flow	$32.0B
Q3 2026 revenue / YoY	$46.72B / +41.2%
Q3 gross margin	66.4% (vs 67.7% Q2)
Advanced node (≤7nm) mix	79% of wafer revenue
HPC / smartphone mix	68% / 21%
Cash and equivalents	$88.2B
Total debt	$32.9B
Net cash	$55.3B
Current ratio	2.46
Quarterly dividend (annualized)	$1.0854 ($4.34, ~1.0% yield)
52-week range	$266.82 – $487.47
Beta (5Y, decaying to 1.05)	1.37

Guidance

Q4 2026 revenue: $52.3-54.3B ACCELERATING

Midpoint $53.3B implies roughly 14% sequential growth, confirming the AI hardware cycle remains in its most aggressive phase. Annualized, this pace supports our FY2026E revenue estimate of $176B.

Q4 2026 gross margin: 65-67% DECELERATING

N2 depreciation continues to suppress peak margins. We expect normalization toward 68%+ as N2 yields mature through 2027.

FY2026 capex: $60-64B STABLE

With $40.64B deployed through nine months, the full-year range implies maximum-velocity capacity adds for CoWoS and N2. This validates our 33% base-case capex assumption.

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Reverse DCF: what the market is pricing in

At $419.06, TSMC carries a $2.17 trillion market cap and a $2.12 trillion enterprise value (net of $55.3B net cash). Holding our base-case FCF margin flat, the market-implied five-year revenue CAGR is **28.7%**, above even our 21% bull case. The verdict is euphoric: the price assumes the AI boom never decelerates and the robotics wave arrives on schedule.

SCENARIO	5-YR REVENUE CAGR	TARGET
Market-implied	28.7%	$419.06 (spot)
Bull case	21%	**$293**
Base case	15%	**$243**
Bear case	9%	**$200**

Model

https://docs.google.com/spreadsheets/d/1QkSrCtmXd20sjRfFtUKPtoBsqs5tDq8LB4IezniCnvM/edit?usp=drivesdk

Five-year detailed EBIT bridge (Revenue to EBIT margin to EBIT to cash taxes to D&A to capex to change in working capital to unlevered FCF), yearly beta decay 1.37 to 1.05, live CAPM off the market-implied ERP tab, Gordon terminal at 3.0%. Blended 25/50/25.

Risks to our Hold

- **To the upside:** AI capex sustains 25%+ growth for five years; robotics silicon orders arrive earlier and larger than modeled; Arizona cost premiums prove smaller than feared; geopolitical risk premium compresses.
- **To the downside:** synchronized hyperscaler capex pause; N2 yields disappoint materially; Taiwan risk reprices sharply; smartphone-plus-PC weakness broadens into HPC.

Live-target disclaimer: price targets in this note are outputs of a live model. The sheet's price, WACC inputs (10-year yield, implied ERP, beta), and share count update with market data, so targets move as inputs move. Targets shown were computed October 15, 2026, at a $419.06 spot. This note is not investment advice and is for entertainment purposes only.

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